

Project Title	 Brain Tumor MRI Image Classification
Skills Takeaway From This Project	• Deep Learning • Python • TensorFlow/Keras or PyTorch • Data Preprocessing • Transfer Learning • Model Evaluation • Streamlit Deployment
Domain	Medical Imaging — Image Classification

Problem Statement

This project aims to develop a deep learning-based solution for classifying brain MRI images into multiple categories according to tumor type. It involves building a custom CNN model from scratch and enhancing performance through transfer learning using pretrained models. The project also includes deploying a user-friendly Streamlit web application to enable real-time tumor type predictions from uploaded MRI images.

Real-Time Business Use Cases

1. AI-Assisted Medical Diagnosis:

Provide radiologists with AI-powered tools to quickly classify brain tumors based on MRI images, reducing diagnostic turnaround time and increasing accuracy.

2. Early Detection and Patient Triage:

Automatically flag high-risk MRI images for immediate specialist review, improving hospital workflow and patient care prioritization.

3. Research and Clinical Trials:

Use AI classification tools to segment patient datasets by tumor type, aiding in research studies and clinical trial recruitment.

4. Second-Opinion AI Systems:

Deploy AI-powered classification tools in telemedicine or remote consultation setups for second-opinion diagnostics in under-resourced healthcare regions.



1. Understand the Dataset

- Review the number of categories (tumor types) and sample images.
- Check for class imbalance and image resolution consistency.
- Explore image distributions visually.

2. Data Preprocessing

- Normalize pixel values to a 0–1 range.
 - Resize images to a consistent shape suitable for model input (e.g. 224x224 pixels).
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3. Data Augmentation

- Apply transformations like rotation, horizontal/vertical flipping, zoom, brightness adjustments, and shifts to artificially increase training data and improve model generalization.
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4. Model Building

- **Custom CNN:** Design a convolutional neural network from scratch, selecting appropriate convolution, pooling, and dense layers.
 - Implement dropout and batch normalization layers to avoid overfitting and stabilize learning.
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5. Transfer Learning

- Load pretrained models (**Example** : ResNet50, MobileNet, InceptionV3, EfficientNetB0) with ImageNet weights.
 - Replace the top classification layers with new dense layers suited for the tumor categories.
 - Optionally unfreeze top layers for fine-tuning after initial training.
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6. Model Training

- Train both custom CNN and transfer learning models.

- Use callbacks like EarlyStopping and ModelCheckpoint to monitor validation loss and save the best performing models.
 - Track training and validation metrics.
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7. Model Evaluation

- Evaluate models using metrics like accuracy, precision, recall, F1-score, and confusion matrix.
 - Visualize model performance trends using training history plots for accuracy and loss.
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8. Model Comparison

- Compare results of custom **CNN** vs **pretrained** models.
 - Identify the most accurate, efficient, and reliable model for deployment.
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9. Streamlit Application Deployment

- Build an interactive web application where users can upload brain MRI images.
 - Display predicted tumor type along with model confidence scores.
 - Ensure the UI is intuitive and informative.
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Dataset

- **Source:**  [Tumour](#) (updated)

Project Deliverables

1. Trained models: **custom CNN** and **pretrained models** (.h5).
2. Streamlit application for tumor classification.
3. Python scripts or notebooks for training, evaluation, and deployment.
4. Model comparison
5. Public GitHub repository with README.
6. Maintain clean, modular, and well-commented code.

Technical Tags

Deep Learning, Image Classification, Medical Imaging, Brain MRI Analysis, CNN, Transfer Learning, TensorFlow, Keras, PyTorch, Data Augmentation, Data Preprocessing, Model Evaluation, Performance Metrics, Streamlit Deployment, Confusion Matrix, Accuracy & Loss Visualization, Model Comparison, Healthcare AI, Computer Vision, Deployment Ready Applications, AI in Radiology

Timeline

The project should be completed and submitted **within 7 days** from the date it is assigned.

Reference

Streamlit recording (English)	 Special session for STREAMLIT(11/08/2024)
Streamlit Reference doc	Streamlit API reference
Project Live Evaluation	 Project Live Evaluation
Capstone Explanation Guideline	 Capstone Explanation Guideline
GitHub Reference	 How to Use GitHub.pptx
Deep learning material	 Deep_Learning-study_material.pdf
Streamlit in colab	 colab_streamlit.ipynb
Project orientation	 Recording : Brain Tumor MRI Image Cl...

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PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Note: You can join the doubt session using the link provided in sheet given below

Timing: Monday-Saturday (3:00PM to 4:00PM)

Booking link : [Project Doubt Session \[DS/AIML\]](#)

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing: Monday-Saturday(11:00AM to 12:00PM) (05:00PM to 06:00PM)

Booking link : <https://forms.gle/1m2Gsro41fLtZurRA>
